YIDI ZHAO

Ph.D. Candidate, Department of Economics, Penn State University

Email: ymz5436@psu.edu

Phone: 814-699-0786

RESEARCH INTERESTS

Econometrics: Model Misspecification, Measurement Error, Statistical Decision Theory
Applied Econometrics: IO, Health, Labor

EDUCATION

Ph.D. in Economics, Department of Economics, Penn State University *May 2027 (expected)*

M.A. in Economics, National School of Development, Peking University *Jul. 2020*

B.A. in Economics, School of Finance, Central University of Finance and Economics *Jul. 2017*

RESEARCH

Job Market Paper

Robust Inference under Local Distribution Shift in Wasserstein Distance

Abstract: This paper studies the robustness of estimators to small departures of the true data-generating process from a reference model that defines the parameter of interest. We measure distributional departures using the quadratic Wasserstein distance, which captures movements of the observations in the sample space due to various factors like measurement errors. We consider local deviations within shrinking Wasserstein neighborhoods so that both sampling uncertainty and distributional bias matter asymptotically. Within this framework, we derive local asymptotic minimax bounds and construct estimators that attain the optimal trade-off between efficiency and robustness. Our results cover both regular parametric models and models defined by unconditional moment restrictions.

Work in Progress

Multinomial Logistic Regression with Sample Selection *Joint work with Onil Boussim (Cal Poly SLO)*

Nonparametric Tests for Proprietary Information at First-Price, Common Value Auctions

TEACHING

Instructor, Department of Economics, Penn State University

- ECON 106: Statistical Foundations for Econometrics (UG) *Summer 2026*

Teaching Assistant, Department of Economics, Penn State University *Fall 2021 – Spring 2025*

- ECON 454: Economics of Mergers (UG); ECON 402: Decision Making and Strategy in Economics (UG); MA ECON 510: Econometrics (Grad)

Teaching Assistant, National School of Development, Peking University

- Advanced Microeconomics II (Graduate core course) *Spring 2019*
- Social Science Survey Data Analysis (UG) *Fall 2018*

SKILLS

Programming: Stata, Python, and Julia.
Language: English (fluent), Chinese (native).

REFERENCES

Marc Henry
Professor of Economics
Penn State University
marc.henry@psu.edu

Andrés Aradillas-López
Professor of Economics
Penn State University
aza12@psu.edu

Keisuke Hirano
Professor of Economics
Penn State University
kuh237@psu.edu